of priority queues. They concluded that the best MST algorithm is Jarník's [7] (also attributed to Prim and Dijkstra; see [3, 7, 12]) as implemented with a pairing heap [13]. Katriel, Sanders, and Träff [10] designed and implemented a non-greedy randomized MST algorithm based on that of Karger et al. [9]. They concluded that on moderately dense graphs, it runs substantially faster than the greedy algorithms tested by Moret and Shapiro.

Cross-References

▸ Randomized Minimum Spanning Tree

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Minimum Weight Triangulation

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Keywords

Minimum length triangulation

Years and Authors of Summarized Original Work

1998; Levcopoulos, Krznaric

Problem Definition

Given a set S of n points in the Euclidean plane, a *triangulation* T of S is a maximal set of nonintersecting straight-line segments whose

endpoints are in S. The *weight* of T is defined as the total Euclidean length of all edges in T. A triangulation that achieves minimum weight is called a *minimum weight triangulation*, often abbreviated MWT, of S.

Key Results

Since there is a very large number of papers and results dealing with minimum weight triangulation, only relatively very few of them can be mentioned here.

Mulzer and Rote have shown that MWT is NP-hard [12]. Their proof of NP-completeness is not given explicitly; it relies on extensive calculations which they performed with a computer. Remy and Steger have shown a quasi-polynomial time approximation scheme for MWT [13]. These results are stated in the following theorem:

Theorem 1 *The problem of computing the MWT (minimum weight triangulation) of an input set S of n points in the plane is NP-hard. However, for any constant $\epsilon > 0$, a triangulation of S achieving the approximation ratio of $1 + \epsilon$, for an arbitrarily small positive constant ϵ, can be computed in time $n^{O(\log^8 n)}$.*

The complexity status of the symmetric problem of finding the *maximum* weight triangulation is still open, but there exists a quasi-polynomial time approximation scheme for it [10].

The Quasi-Greedy Triangulation Approximates the MWT

Levcopoulos and Krznaric showed that a triangulation of total length within a constant factor of MWT can be computed in polynomial time for arbitrary point sets [7]. The triangulation achieving this result is a modification of the so-called *greedy* triangulation. The greedy triangulation starts with the empty set of diagonals and keeps adding a shortest diagonal not intersecting the diagonals which have already been added, until a full triangulation is produced. The greedy triangulation has been shown to approximate the minimum weight triangulation within a constant factor, unless a special case arises where the greedy diagonals inserted are "climbing" in a special, very unbalanced way along a relatively long concave chain containing many vertices and with a large empty space in front of it, at the same time blocking visibility from another, opposite concave chain of many vertices. In such "bad" cases, the worst-case ratio between the length of the greedy and the length of the minimum weight triangulation is shown to be $\Theta\left(\sqrt{n}\right)$. To obtain a triangulation which always approximates the MWT within a constant factor, it suffices to take care of this special bad case in order to avoid the unbalanced "climbing," and replace it by a more balanced climbing along these two opposite chains. Each edge inserted in this modified method is still almost as short as the shortest diagonal, within a factor smaller than 1.2. Therefore, the modified triangulation which always approximates the MWT is named the *quasi-greedy* triangulation. In a similar way as the original greedy triangulation, the quasi-greedy triangulation can be computed in time $O(n \log n)$ [8]. Gudmundsson and Levcopoulos [5] showed later that a variant of this method can also be parallelized, thus achieving a constant factor approximation of MWT in $O(\log n)$ time, using $O(n)$ processors in the CRCW PRAM model. Another by-product of the quasi-greedy triangulation is that one can easily select in linear time a subset of its edges to obtain a convex partition which is within a constant factor of the minimum length convex partition of the input point set. This last property was crucial in the proof that the quasi-greedy triangulation approximates the MWT. The proof also uses an older result that the (original, unmodified) greedy triangulation of any convex polygon approximates the minimum weight triangulation [9]. Some of the results from [7] and from [8] can be summarized in the following theorem:

Theorem 2 *Let S be an input set of n points in the plane. The quasi-greedy triangulation of S, which is a slightly modified version of the greedy triangulation of S, has total length within a constant factor of the length of the MWT (minimum weight triangulation) of S and can be computed in time $O(n \log n)$. Moreover, the*

(unmodified) greedy triangulation of S has length within $O\left(\sqrt{n}\right)$ of the length of MWT of S, and this bound is asymptotically tight in the worst case.

Computing the Exact Minimum Weight Triangulation

Below, three approaches to compute the exact MWT are shortly discussed. These approaches assume that it is numerically possible to efficiently compare the total length of sets of line segments in order to select the set of smallest weight. This is a simplifying assumption, since this is an open problem per se. However, the problem of computing the exact MWT remains NP-hard even under this assumption [12]. The three approaches differ with respect to the creation and selection of subproblems, which are then solved by dynamic programming.

The first approach, sketched by Lingas [11], employs a general method for computing optimal subgraphs of the complete Euclidean graph. By developing this approach, it is possible to achieve subexponential time $2^{O(\sqrt{n}\log n)}$. The idea is to create the subproblems which are solved by dynamic programming. This is done by trying all (suitable) planar separators of length $O\left(\sqrt{n}\right)$, separating the input point set in a balanced way, and then to proceed recursively within the resulting subproblems.

The second approach uses fixed-parameter algorithms. So, for example, if there are only $O(\log n)$ points in the interior of the convex hull of S, then the MWT of S can be computed in polynomial time [4]. This approach extends also to compute the minimum weight triangulation under the constraint that the outer boundary is not necessarily the convex hull of the input vertices; it can be an arbitrary polygon. Some of these algorithms have been implemented; see Grantson et al. [2] for a comparison of some implementations. These dynamic programming approaches take typically cubic time with respect to the points of the boundary but exponential time with respect to the number of remaining points. So, for example, if k is the number of hole points inside the boundary polygon, then an algorithm, which has also been implemented, can compute the exact MWT in time $O(n^3 \cdot 2^k \cdot k)$ [2].

In an attempt to solve larger problems, a different approach uses properties of MWT which usually help to identify, for random point sets, many edges that *must* be, respectively *cannot* be, in MWT. One can then use dynamic programming to fill in the remaining MWT edges. For random sets consisting of tens of thousands of points from the uniform distribution, one can thus compute the exact MWT in minutes [1].

Applications

The problem of computing a triangulation arises, for example, in finite element analysis, terrain modeling, stock cutting, and numerical approximation [3, 6]. The *minimum weight* triangulation has attracted the attention of many researchers, mainly due to its natural definition of optimality, and because it has proved to be a challenging problem over the past 30 years, with unknown complexity status until the end of 2005.

Open Problems

All results mentioned leave open problems. For example, can one find a simpler proof of NP-completeness, which can be checked without running computer programs? It would be desirable to improve the approximation constant which can be achieved in polynomial time (to simplify the proof, the constant shown in [7] is not explicitly calculated and it would be relatively large, if the proof is not refined). The time bound for the approximation scheme could hopefully be improved. It could also be possible to refine the software which computes efficiently the exact MWT for large random point sets, so that it can handle efficiently a wider range of input, i.e., not only completely random point sets. This could perhaps be done by combining this software with implementations of fixed-parameter algorithms, as the ones reported in [2, 4], or with other approaches. It is also open whether or not the

subexponential exact method can be further improved.

Experimental Results

Please see the last paragraph under the section about key results.

URL to Code

Link to code used to compare some dynamic programming approaches in [2]: http://fuzzy.cs.unimagdeburg.de/~borgelt/pointgon.html

Cross-References

▸ Greedy Set-Cover Algorithms
▸ Minimum Geometric Spanning Trees
▸ Minimum k-Connected Geometric Networks

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Minimum Weighted Completion Time

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